R9 - 29

TEZPUR UNIVERSITY

TU/CSE

Spring Semester Mid Term (Test II) Examination, 2022

CO218: DATA COMMUNICATION

Full Marks: 40 *Time: 2 hrs*

All questions are compulsory. Numbers in right hand side indicates marks. Write the answers precise.

1. Answer to the questions below adequately (**answer any 5**) [2x5=10]
   (a) Difference between Thermal noise and Intermodulation noise
   (b) What is power spectral density (PSD) in reference to bandwidth of a channel?
   (c) Name the transport protocols implemented by Novel NETWARE
   (d) How do you differentiate SSB from DSB-SC for amplitude modulation?
   (e) "Given a bandwidth of B, the highest signal rate that can be carried is 2B" Justify the statement.
   (f) What do you understand by ECC and EDC in context of error control?

2. Define different types of crosstalk impairments faced by a communication system. What is the reason of *aliasing* impairment that may arise in a communication system? [3+1=4]

3. State the functionalities of Data Link and Network Layers of ISO-OSI protocol stack. Where do we implement end-to-end transport service and why? [3+2=5]

4. Which mode of optical fibre transmission suffers the issue of "spacing between pulses limits data rate"? Explain briefly in what situation this happens. [1+2=3]

5. Consider a carrier signal needs to carry user's data in which one data element is encoded as two signal element with the bit rate 10 kbps. Calculate the average value of the baud rate if case factor remains 0.5. [4]

   Or

   Consider the vacuum space wavelength of an optical fibre supporting the single mode transmission is cited as 1550 nm. If the velocity of light propagation on it is 2.04×10^8 m/s, calculate the actual wavelength that can be attainable on the fibre.

6. What is the capacity of a channel with bandwidth 2 MHz for an encoding scheme with 32 signal levels? What signal-to-noise ratio is required to achieve the above calculated capacity? [3+3=6]

7. If the received signal level for a particular digital system is -151dBW and the receiver system effective noise temperature is 1500^0 K, what is the E_b/N_o for a link transmitting at 2400 bps? [4]

8. A digital signaling system is required to operate at a capacity of 8 Mbps. If the fundamental frequency of the carrier signal is 2 MHz, what is the minimum required bandwidth of the channel? [4]

========